Supplementary Material (Online Resource 1)

Real-Time Multi-Behavior Detection of Group-Housed Pigs on Edge Devices: Class-Imbalance-Aware Sampling and a Lightweight FasterNet Backbone — Supplementary Material

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Table S1. Per-class AP50 on the validation set. On the held-out test set, the key rare-class result is active 0.526 → 0.639 (baseline → M4, +11.3 points); sitting 0.534 → 0.571; fight 0.840 → 0.871.

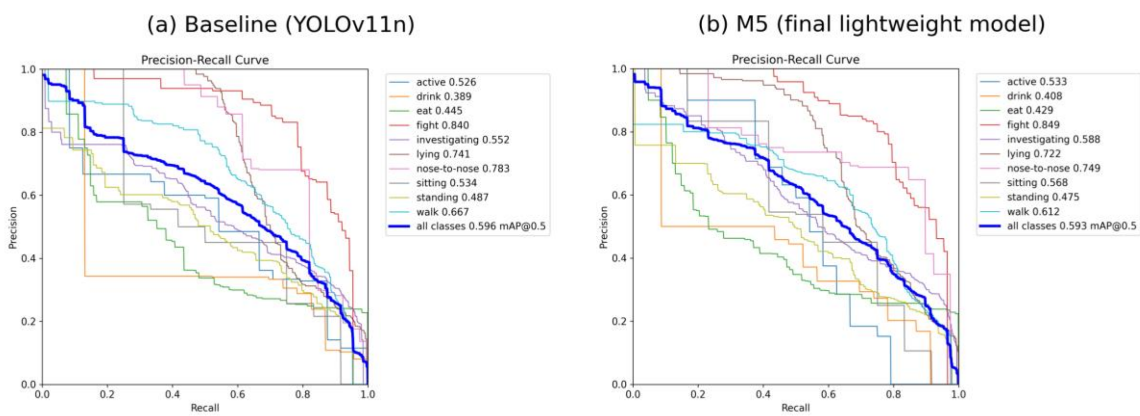

Fig. S1 Precision–recall curves on the held-out test set for the baseline (left) and the final lightweight model M5 (right). Behaviour-class curves are shown thin; the thick blue curve is the all-class average (mAP50 in the legend)

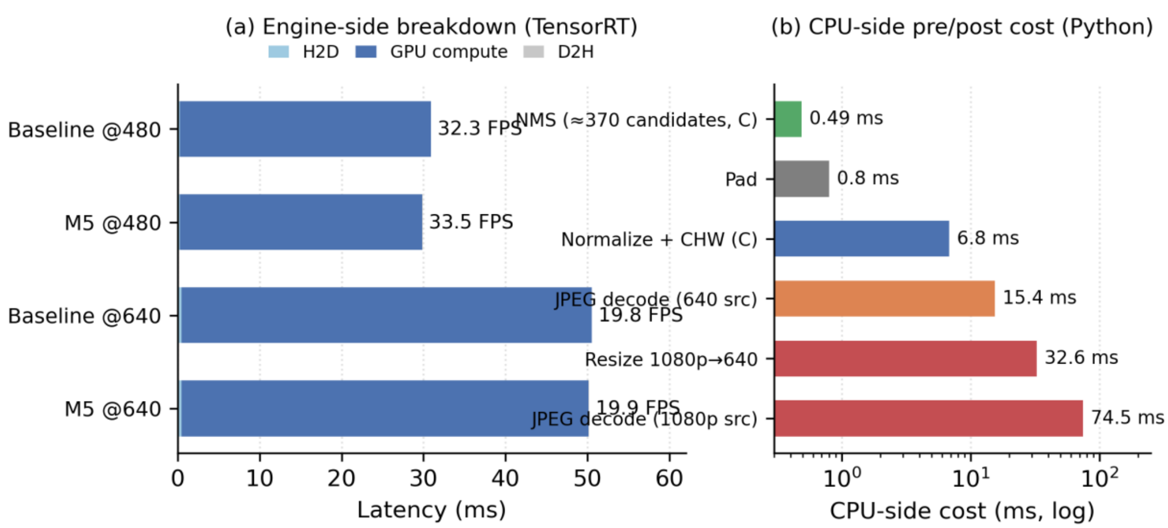

Fig. S2 Latency budget on the Jetson Nano. (a) Engine-side decomposition (TensorRT measurement): GPU compute dominates at both input sizes; H2D/D2H transfers are negligible. (b) CPU-side pre/post-processing costs in a pure-Python pipeline (log scale): naive decode and resize can exceed GPU compute, so production pipelines should implement pre/post-processing in C or CUDA

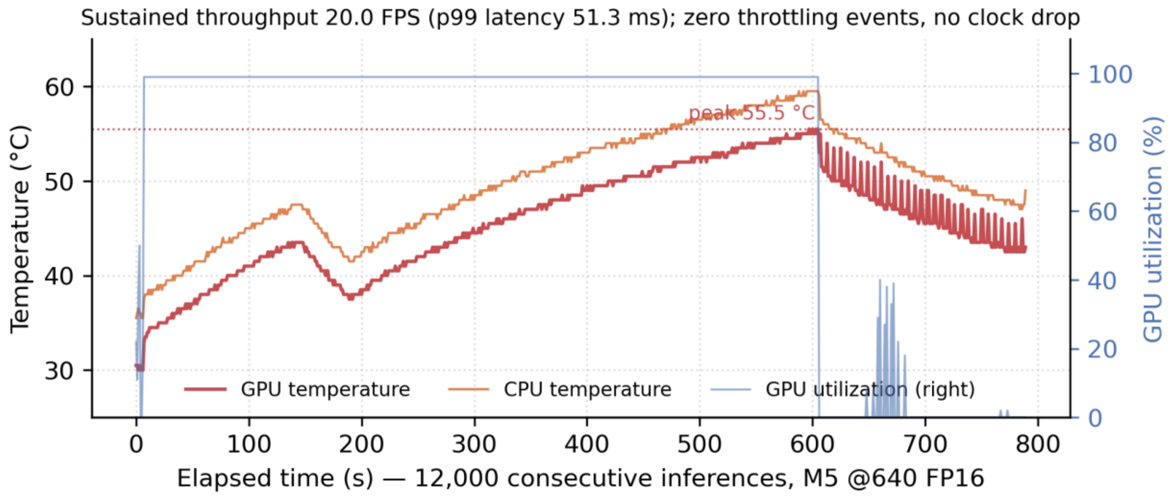

Fig. S3 Sustained-operation timeline on the Jetson Nano over 12,000 consecutive inferences (10.7 min, M5 at 640×640 FP16). Throughput holds at 20.0 FPS (99th-percentile latency 51.3 ms); GPU temperature peaks at 55.5 °C with zero thermal-throttling events and no clock drop

Class	Baseline	M3 (FasterNet)	M4 (sampling)	M5 (combined)	M4 – baseline
active	0.459	0.372	0.552	0.452	+9.3
drink	0.408	0.440	0.459	0.430	+5.1
eat	0.436	0.437	0.447	0.449	+1.1
fight	0.858	0.828	0.842	0.786	−1.6
investigating	0.581	0.571	0.570	0.577	−1.1
lying	0.783	0.764	0.766	0.743	−1.7
nose-to-nose	0.686	0.655	0.622	0.710	−6.4
sitting	0.403	0.233	0.423	0.349	+2.0
standing	0.438	0.429	0.449	0.445	+1.1
walk	0.687	0.646	0.686	0.667	−0.1
mAP50 (all)	0.573	0.537	0.582	0.561	+0.9

Table S2. Per-class AP50 on the held-out test set for the baseline and the two adopted variants. The rare-class pattern mirrors the validation set (Table S1): the gains of M4 concentrate on active, sitting and fight; M5 keeps part of the rare-class gain while carrying 4.4% fewer parameters. All entries are single-run values (seed 0); the three-seed means for the overall metric are reported in Section 4.5 of the main text. M5 values were re-measured under the official validation protocol (overall mAP50 0.5933, matching the cloud-reported 0.5932).

Class	Baseline	M4 (sampling)	M5 (combined)
active	0.526	0.639	0.533
drink	0.389	0.373	0.408

eat	0.445	0.418	0.429
fight	0.840	0.871	0.849
investigating	0.552	0.562	0.588
lying	0.741	0.737	0.722
nose-to-nose	0.783	0.740	0.749
sitting	0.534	0.571	0.568
standing	0.487	0.468	0.475
walk	0.667	0.655	0.612
mAP50 (all)	0.596	0.604	0.593